Types of Scientific Research

(Biostatistics and Scientific Research)

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Week-1

Outlines

- Basic research and Applied research
- Quantitative Research
- Qualitative Research
- Mixed Research
- Other Types of Scientific Research
- Online Scientific Research
- Scientific Research Engines
- Database vs Publisher

Types of Scientific Research

Basic research

Solve a problem that hasn't any application to any practical problem in the world e.g. fill the gap

Applied research

When the solution to a research problem does have practical consequences, e.g., a disease needs therapy

Types of Scientific Research

Quantitative Research

- Studies are based on counting or **measuring** some type of data.
- Aim to evaluate the "**when**," "**where**," and "**what**" of the research topic or question.
- It basically measures **numerical data** to discover patterns.

Qualitative Research

- It focuses on the quality of the data rather than the quantity.
- Aim to investigate the "**why**" and the "**how**" of the decision-making process.
- It leads to a **solution** after a better understanding of the problem.

Quantitative research

Definition: Quantitative research involves the collection and analysis of **numerical data** to answer research questions or test hypotheses

Methodology: involves **surveys**, **experiments**, and **observational studies**

Data Collection: Data is collected using **structured instruments**, such as questionnaires or surveys

Data Analysis: Data is analyzed using **statistical** methods, such as regression analysis or hypothesis testing.

Outcome: The outcome of quantitative research is often **a numerical value** or a statistical model

Qualitative research

Definition: Qualitative research involves the collection and analysis of **non-numerical data**, such as text, images, or observations, to gain a deeper understanding of a phenomenon

Methodology: involves case **studies**, **content analysis**, and ethnographic research

Data Collection: Data is collected using **unstructured or semi-structured** instruments, such as interviews or focus groups

Data Analysis: Data is analyzed using **thematic analysis**, content analysis, or other qualitative methods

Outcome: The outcome of qualitative research is often a rich, detailed **description** of a phenomenon or a set of themes or patterns.

Mixed Research

- **Definition:** It combines **both quantitative and qualitative** research methods to achieve a more comprehensive understanding of a research problem.

- **Characteristics of Mixed Research:**

1. Mixed research **combines** the strengths of both quantitative and qualitative research methods

2. **Integration** of data collection and analysis methods from both quantitative and qualitative traditions

3. **Flexibility** in research design: it allows for flexibility in research design, enabling researchers to adapt to changing research conditions

4. **Comprehensive understanding**: Mixed research aims to provide a more comprehensive understanding of a research problem or phenomenon.

Other Types of Scientific Research

1. Exploratory Research
2. Descriptive research
3. Explanatory research
4. Longitudinal Research
5. Cross sectional Research
6. Action research

Online Scientific Research

https://scholar.google.com.eg/

https://www.ncbi.nlm.nih.gov/pubmed/

https://www.scopus.com/home.uri

https://www.sciencedirect.com/

https://www.ekb.eg/

https://www.frontiersin.org/

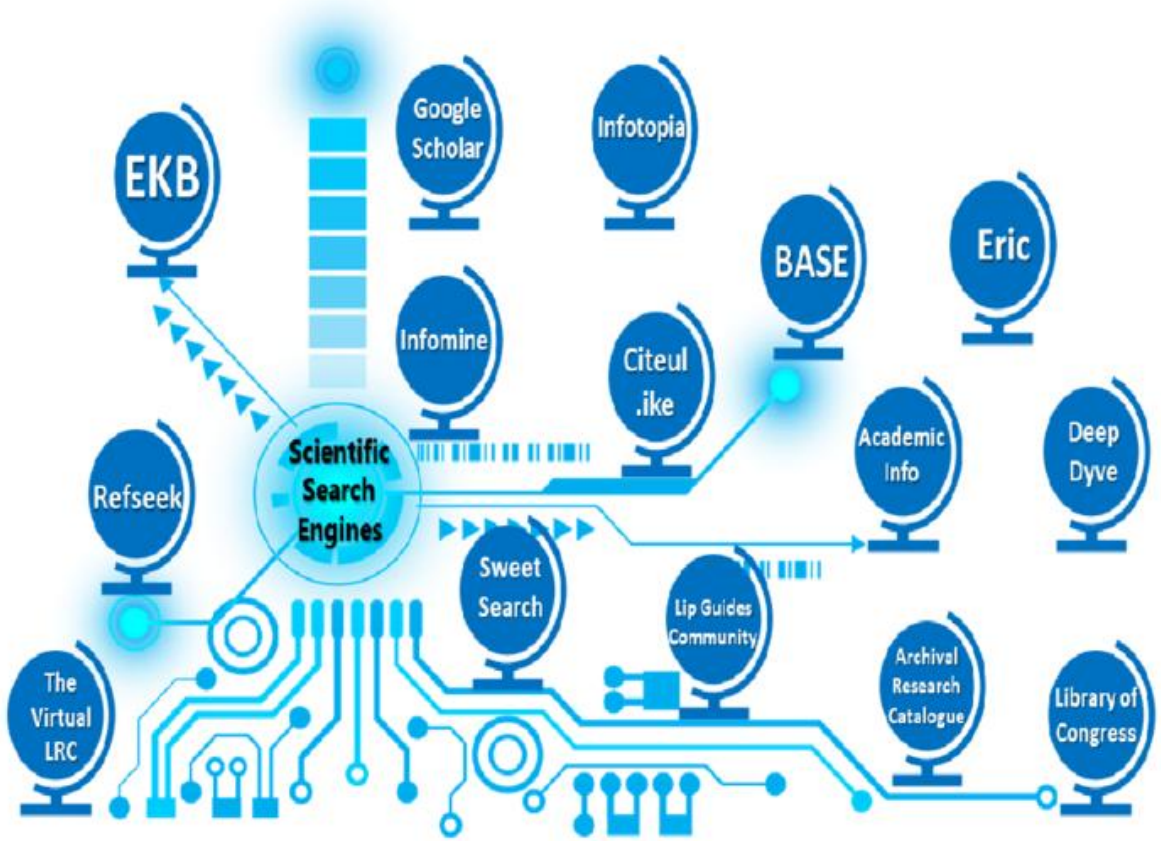
https://www.nature.com/siteindex

https://www.springer.com/gp

https://www.wiley.com/en-us

https://www.hindawi.com/

Scientific Research Engines and databases



Database vs Publisher

- **Publisher:** an organization that publishes books, magazines, or newspapers
- **Database:** is an online tool comprised of many journals that often focus on a broad area
- Databases are constantly updated
- The database's name doesn't always indicate its scope

Paper < Journal < Database < Publisher

Most Famous Online Scientific Publishers

- Springer
- Cambridge university press
- John Wiley & Sons
- Blackwell Publishing
- SAGE Publishing
- Taylor & Francis
- MDPI
- Elsevier
- Nature

Most common databases

Content:

1. Scopus

2. Web of Science

3. PubMed

4. ERIC

5. IEEE Xplore

6. ScienceDirect

7. Directory of Open Access Journals (DOAJ)

8. JSTOR

Scientific written communication

You can use any of these ways to communicate in a scientific way according to your aim, audience.

- Journal articles
- Review article
- Book reviews and book chapters
- Research or grant proposals
- Posters
- Clinical case study
- Clinical trial
- Perspective and Opinion
- Theses or dissertations
- Slide presentations
- Letter to the editor
- Reports

THANK
YOU